

Construction Site Management System

Requirement gathering/analysis:

Site creation & management

Multiple construction site handling

Worker account creation

Worker details view & management

Daily worker allocation to sites (based on availability/skill)

Site work status tracking

- Pending
- In Progress
- Completed
- Deadline-based tracking

Worker attendance tracking (daily basis)

Weekly salary calculation

- Based on working days
- Based on worker type (skilled / unskilled)

Dashboard overview

- Total sites
- Active workers
- Work status summary

Report generation

- Site-wise reports
- Worker-wise reports

Notification system

- Daily work assignment alerts

Authentication system

- Login (Owner / Supervisor roles)

Phase 1 :

- Authentication
 - Login (Owner, Supervisor)
- Site Management
 - Create site
 - View site details
 - Update site info
- Employee Management
 - Add worker
 - View worker details
 - Track attendance
- Task / Work Management
 - Assign daily work
 - Track work status
 - Pending / In Progress / Completed
- Daily Allocation
 - Assign workers to sites
 - Handle deadline-based priority
- Work Status Tracking
 - Update & monitor site progress
- Salary Calculation
 - Calculate weekly salary
 - Based on attendance + worker type

Model Classes :

Users :

```
int id;          // PRIMARY KEY AUTO_INCREMENT
String name
String username
String password
String role; // Owner/Supervisor
```

Sites :

```
int id          // PRIMARY KEY AUTO_INCREMENT
String name
String location
status:Enum - InProgress/Completed
Long startDate
Long endDate
```

Workers :

```
int id          // PRIMARY KEY AUTO_INCREMENT
String name
String type // Electrician / Mason etc
double dailyWage
```

Attendance :

```
int id          // PRIMARY KEY AUTO_INCREMENT
int workerId    //FOREIGN KEY (worker_id) REFERENCES
```

workers(id)

```
Long date
boolean present
```

Task :

```
int id
int siteId      //FOREIGN KEY (site_id) REFERENCES sites(id)
int workerId    //FOREIGN KEY (worker_id) REFERENCES
```

workers(id)

String description
Status: Enum- Pending / In Progress / Completed
Long assignedDate
Long deadline

Allocation :

int id
int workerId //FOREIGN KEY (worker_id) REFERENCES
workers(id)
int siteId //FOREIGN KEY (site_id) REFERENCES sites(id)
Long date

Salary

int id
int workerId //FOREIGN KEY (worker_id) REFERENCES
workers(id)
int totalDays
double totalAmount
Long weekStart
Long weekEnd

GitHub Link : <https://github.com/gowtham270606/ConstructionHub.git>